Schrödinger Bridge Benchmark Construction

1. Dataset construction with known Schrödinger Bridge

Let $\Omega = \mathcal{C}([0,1], \mathbb{R}^d)$ be the path space and let $\mathcal{P}(\Omega)$ denote the space of probability measures on Ω. Let $R \in \mathcal{P}(\Omega)$ be the reference measure of the Schrödinger Bridge problem, defined as the law of the stochastic process $(X_t)_{t\in[0,1]}$ governed by the following linear SDE:

$$\mathrm{d}X_t = (\alpha(t)X_t + \beta(t))\mathrm{d}t + \sigma(t)\mathrm{d}W_t \tag{1}$$

where $\alpha : [0,1] \to \mathbb{R}$, $\beta : [0,1] \to \mathbb{R}^d$ and $\sigma : [0,1] \to (0,\infty)$ are deterministic, continuous functions.

We utilize the methodology from the paper Building the Bridge of Schrödinger: A Continuous Entropic Optimal Transport Benchmark and adapt it to the Schrödinger Bridge problem with an arbitrary linear reference measure. This adaptation relies on the dual formulation detailed in the notes (page 11,(page 11, *Apply EBM to the Schrödinger Bridge Problem*).

Let $\mathcal{X}$ and $\mathcal{Y}$ be Polish spaces, and let $\pi^* \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ denote the optimal coupling, i.e., the solution of the static Schrödinger Bridge problem with respect to marginals $\mu \in \mathcal{P}(\mathcal{X})$ and $\nu \in \mathcal{P}(\mathcal{Y})$, and reference measure R. Let $\phi^* : \mathcal{X} \to \mathbb{R}$ and $\psi^* : \mathcal{Y} \to \mathbb{R}$ denote the Schrödinger potentials, i.e., the solutions of the dual Schrödinger Bridge problem. Then, it can be shown that the relation between the optimal coupling and the potentials is given by:

$$\mathrm{d}\pi^*(x,y) = \frac{\exp(\phi^*(x) + \psi^*(y))}{\int_{\mathcal{X}\times\mathcal{Y}} \exp(\phi^*(x) + \psi^*(y))\mathrm{d}R_{01}(x,y)}\mathrm{d}R_{01}(x,y)$$

where $R_{01}(\cdot,\cdot) = R(X_0 \in \cdot, X_1 \in \cdot)$.

By the disintegration theorem, the joint measure R_{01} can be factored into a marginal measure $R_0 \in \mathcal{P}(\mathcal{X})$ and a markov transition kernel $R_{01}(\cdot \mid x)$ on $\mathcal{Y}$. In differential notation, this is commonly written as:

$$\mathrm{d}R_{01}(x,y) = R_{01}(\mathrm{d}y \mid x)\mathrm{d}R_0(x)$$

If we consider the same factorization on the optimal coupling π^* then it can be shown that the transition kernel of π^* depends only on the Schrödinger potential ψ^* as follows:

$$\pi^*(\mathrm{d}y \mid x) = \frac{1}{Z_x}\exp(\psi^*(y))R_{01}(\mathrm{d}y \mid x) \tag{2}$$

where Z_x is the partition function defined as $Z_x = \int_{\mathcal{Y}} \exp(\psi^*(y))R_{01}(\mathrm{d}y \mid x)$.

Following the methodology of the aforementioned paper, we will parameterize the exponentiated dual potential $\exp(\psi^*(y))$ as a Gaussian Mixture Model (GMM). By starting

with initial samples $x \sim \mu$ and drawing directly from the conditional plan $\pi^*(\cdot \mid x)$, we generate terminal samples y that implicitly construct the target marginal distribution ν. Before detailing this sampling procedure, we must first derive the closed-form expression for the reference transition kernel $R_{01}(\cdot \mid x)$.

By solving the linear SDE (1), we obtain the explicit continuous-time solution for the state X_t:

$$X_t = X_0 \exp\left(\int_0^t \alpha(u)\mathrm{d}u\right) + \int_0^t \exp\left(\int_s^t \alpha(u)\mathrm{d}u\right)\beta(s)\mathrm{d}s + \int_0^t \exp\left(\int_s^t \alpha(u)\mathrm{d}u\right)\sigma(s)\mathrm{d}W_s$$

Conditioned on the initial state $X_0 = x$, the mean and covariance of this process are given by:

$$\mathbb{E}[X_t \mid X_0 = x] = x \exp\left(\int_0^t \alpha(u)\mathrm{d}u\right) + \int_0^t \exp\left(\int_s^t \alpha(u)\mathrm{d}u\right)\beta(s)\mathrm{d}s,$$
$$\mathrm{Cov}(X_t \mid X_0 = x) = \left(\int_0^t \exp\left(2\int_s^t \alpha(u)\mathrm{d}u\right)\sigma^2(s)\mathrm{d}s\right) I_d.$$

To evaluate this at the terminal time $t = 1$, we introduce the deterministic constants $A_{01} = \exp(\int_0^1 \alpha(u)\mathrm{d}u)$ and $B_{01} = \int_0^1 \exp(\int_s^1 \alpha(u)\mathrm{d}u)\beta(s)\mathrm{d}s$. The conditional mean of the terminal state X_1 given $X_0 = x$ simplifies to:

$$\mu_{01}(x) = A_{01}x + B_{01}$$

Similarly, defining the variance scalar $V_{01} = \int_0^1 \exp(2\int_s^1 \alpha(u)\mathrm{d}u)\sigma^2(s)\mathrm{d}s$, the conditional covariance matrix becomes:

$$\Sigma_{01} = V_{01} I_d$$

Because the original SDE (1) is linear with deterministic coefficients, the conditional distribution of the terminal state is strictly Gaussian:

$$X_1|_{X_0=x} \sim \mathcal{N}(\mu_{01}(x), \Sigma_{01})$$

Consequently, with a slight abuse of notation, we denote the probability density function of the transition kernel $R_{01}(\mathrm{d}y \mid x)$ with respect to the Lebesgue measure simply as $R_{01}(y \mid x)$:

$$R_{0,1}(y \mid x) = \mathcal{N}(y;\ \mu_{01}(x), \Sigma_{01})$$

We also denote the probability density of the transition kernel $\pi^*(\mathrm{d}y \mid x)$ as $\pi^*(y \mid x)$ equation (2) becomes:

$$\pi^*(y \mid x) = \frac{1}{Z_x}\exp(\psi^*(y))\mathcal{N}(y;\ \mu_{01}(x), \Sigma_{01})$$

We then define the Schrödinger potential ψ^* as the logarithm of a GMM, namely:

$$\psi^*(y) = \log\left(\sum_{k=1}^{K} w_k \mathcal{N}(y;\ \mu_k, \Sigma_k)\right)$$

With that definition the conditional density of the optinal coupling is evaluated as follows:

$$\pi^*(y \mid x) = \frac{1}{Z_x} \sum_{k=1}^{K} w_k\, \mathcal{N}(y;\ \mu_k, \Sigma_k)\, \mathcal{N}(y;\ \mu_{01}(x), \Sigma_{01})$$

To derive a closed-form expression for the conditional density $\pi^*(y \mid x)$, we rely on the property that the product of two Gaussian probability density functions yields an unnormalized Gaussian. Specifically, for each component k of the mixture:

$$\mathcal{N}(y;\ \mu_k, \Sigma_k)\, \mathcal{N}(y;\ \mu_{01}(x), \Sigma_{01}) = \mathcal{N}(\mu_{01}(x)\ ; \mu_k, \Sigma_k + \Sigma_{01})\, \mathcal{N}(y\ ; m_k(x), S_k)$$

where the conditional covariance matrix is defined as $S_k = (\Sigma_k^{-1} + \Sigma_{01}^{-1})^{-1}$, and the conditional mean vector is $m_k(x) = S_k(\Sigma_k^{-1}\mu_k + \Sigma_{01}^{-1}\mu_{01}(x))$.

Recall that the partition function evaluates the integral $Z_x = \int_{\mathcal{Y}} \exp(\psi^*(y)) R_{01}(y \mid x)\mathrm{d}y$. Substituting the GMM parameterization for the exponentiated potential and applying the Gaussian product rule derived above, the integral of the y-dependent term $\mathcal{N}(y; m_k(x), S_k)$ evaluates to 1, leaving only the scaling factors:

$$Z_x = \sum_{k=1}^{K} w_k\, \mathcal{N}(\mu_{01}(x);\ \mu_k, \Sigma_k + \Sigma_{01})$$

Let us denote these state-dependent, unnormalized component weights as $c_k(x) = w_k \mathcal{N}(\mu_{01}(x); \mu_k, \Sigma_k + \Sigma_{01})$, and define the normalized weights:

$$\tilde{w}_k(x) = \frac{c_k(x)}{\sum_{k=1}^{K} c_k(x)}$$

Finally, substituting these back into the original formulation yields the closed-form expression for the optimal conditional plan, which is exactly a Gaussian Mixture Model with state-dependent weights and shifted means:

$$\pi^*(y \mid x) = \sum_{k=1}^{K} \tilde{w}_k(x)\, \mathcal{N}(y;\ m_k(x), S_k) \tag{3}$$

Although the reference process and the base transition kernel are built upon strictly linear-Gaussian dynamics, parameterizing the exponentiated dual potential $\exp(\psi^*(y))$

as a Gaussian Mixture Model guarantees that the resulting terminal marginal distribution ν is highly expressive. Because the target distribution is constructed by integrating the optimal conditional plan over the initial prior, $\nu(y) = \int_{\mathcal{X}} \pi^*(y \mid x)\mathrm{d}\mu(x)$, the terminal marginal effectively becomes a continuous, state-dependent mixture of GMMs. Since a standard GMM with a sufficient number of components K can already approximate any smooth probability density to arbitrary precision, this integrated formulation acts as a robust universal approximator for the target data space. Consequently, this benchmark can seamlessly scale from simple unimodal datasets to highly complex, multimodal target marginals ν (such as high-dimensional image manifolds), ensuring the generative modeling task remains realistically challenging.

Algorithm 1: Schrödinger Bridge Benchmark Data Generation

Input: Prior distribution μ, reference SDE coefficients α, β, σ, GMM potential parameters $\{w_k, \mu_k, \Sigma_k\}_{k=1}^K$, sample size N

$$A_{01} = \exp\left(\int_0^1 \alpha(u)\mathrm{d}u\right); \quad B_{01} = \int_0^1 \exp\left(\int_s^1 \alpha(u)\mathrm{d}u\right)\beta(s)\mathrm{d}s$$

$$\Sigma_{01} = \left(\int_0^1 \exp\left(2\int_s^1 \alpha(u)\mathrm{d}u\right)\sigma^2(s)\mathrm{d}s\right) I_d$$

$$S_k = (\Sigma_k^{-1} + \Sigma_{01}^{-1})^{-1} \quad \forall k \in \{1, \ldots, K\}$$

for $i = 1, \ldots, N$ **do**

$$x_i \sim \mu; \quad \mu_{01}(x_i) = A_{01}x_i + B_{01}$$

$$m_k(x_i) = S_k(\Sigma_k^{-1}\mu_k + \Sigma_{01}^{-1}\mu_{01}(x_i)) \quad \forall k \in \{1, \ldots, K\}$$

$$c_k(x_i) = w_k \mathcal{N}(\mu_{01}(x_i);\ \mu_k, \Sigma_k + \Sigma_{01}) \quad \forall k \in \{1, \ldots, K\}$$

$$\tilde{w}_k(x_i) = \frac{c_k(x_i)}{\sum_{j=1}^K c_j(x_i)} \quad \forall k \in \{1, \ldots, K\}$$

$$k^* \sim \text{Categorical}(\tilde{w}_1(x_i), \ldots, \tilde{w}_K(x_i))$$

$$y_i \sim \mathcal{N}(m_{k^*}(x_i), S_{k^*})$$

Output: Ground-truth static optimal coupling dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N \sim \pi^*$

Because the optimal conditional plan $\pi^*(y \mid x)$ takes the form of a Gaussian Mixture Model, generating a target point y_i for a given initial state x_i cannot be done by evaluating the density function directly. Instead, it must be executed via a two-step ancestral sampling procedure, as detailed in Algorithm 1:

1. The Routing Step: We first determine which specific component of the mixture will generate the sample. For a given starting point x_i, each component k is assigned a state-dependent probability $\tilde{w}_k(x_i)$ of being selected. We sample exactly one discrete integer index $k^* \in \{1, \ldots, K\}$ from a Categorical distribution parameterized by these normalized weights.

2. The Generative Step: Once the component index k^* is selected, we conditionally ignore all other components. The continuous terminal point y_i is then drawn strictly from the winning component's multivariate Gaussian distribution: $y_i \sim \mathcal{N}(m_{k^*}(x_i), S_{k^*})$.

This discrete routing step is strictly necessary to preserve the expressivity of the target marginal distribution ν. If one were to incorrectly draw a sample from every component and combine them linearly (e.g., $y_i = \sum_{k=1}^{K} \tilde{w}_k(x_i) y_k$), the benchmark would structurally fail. A fundamental property of probability theory dictates that a linear combination of independent Gaussian random variables strictly yields another simple, unimodal Gaussian. Blending the individual component samples together would mathematically collapse the mixture, completely erasing the complex, multimodal, and non-linear topology that this GMM parameterization is specifically designed to evaluate.

2. Deriving the optimal drift of the GMM parameterization

Let the reference measure R be defined by the stochastic differential equation (SDE) driven by a standard Brownian motion W_t:

$$\mathrm{d}X_t = f(t, X_t)\mathrm{d}t + \sigma(t)\mathrm{d}W_t$$

The Schrödinger Bridge problem seeks an optimal path measure P^* that satisfies specific marginal constraints. In path space, this optimal measure is constructed via a Radon-Nikodym derivative from R, driven by a terminal Schrödinger potential $\psi^*(X_1)$:

$$\frac{\mathrm{d}P^*}{\mathrm{d}R} \propto \exp(\psi^*(X_1))$$

To determine the local dynamics of the optimal process at any intermediate time $t \in [0, 1]$, we must evaluate the "pull" of this terminal constraint. We define the space-time harmonic function $h(x, t)$ as the conditional expectation of the terminal boundary under the unsteered reference process:

$$h(t, x) = \mathbb{E}_R[\exp(\psi^*(X_1)) \mid X_t = x]$$

Because $h(t, X_t)$ is a conditional expectation under the Markov process R, it is strictly a martingale. According to Doob's h-transform, conditioning a measure on a martingale leaves the underlying volatility $\sigma(t)$ completely unaltered but shifts the drift.

By applying Itô's lemma to $h(t, X_t)$ and utilizing Girsanov's theorem, this change of measure translates into an exact drift correction proportional to the spatial gradient of the log-martingale. If we define our time-dependent Schrödinger potential as the

logarithm of this harmonic function, $\psi(t,x) = \log h(t,x)$, the Girsanov drift correction for the arbitrary reference drift $f(t,x)$ becomes:

$$v^*(t,x) = f(t,x) + \sigma^2(t)\nabla_x \psi(t,x)$$

This specific drift is guaranteed to be the optimal continuous-time control policy. Because the harmonic function $h(t,x)$ is constructed directly from the terminal Schrödinger potential ψ^*, the resulting path measure is the unique minimizer of the Kullback-Leibler divergence with respect to the reference measure R. Consequently, $v^*(t,x)$ represents the exact optimal vector field that steers the initial distribution to the target distribution with minimal entropic cost.

We now apply this general framework to our specific benchmark construction. Recall that our reference measure R is governed by the linear SDE with drift $f(x,t) = \alpha(t)x + \beta(t)$. To compute the harmonic function $h(x,t)$, we must evaluate the conditional expectation of the exponentiated terminal potential under this reference process:

$$h(x,t) = \int_{\mathcal{Y}} \exp(\psi^*(y)) R_{t1}(y \mid x) \mathrm{d}y$$

where $R_{t1}(y \mid x)$ is the transition kernel of the reference SDE from an intermediate time t to the terminal time $t = 1$. Because the SDE is strictly linear, this transition kernel is purely Gaussian: $R_{t1}(y \mid x) = \mathcal{N}(y;\ \mu_{t1}(x), \Sigma_{t1})$. By integrating the SDE dynamics from t to 1, the conditional mean and covariance evaluate to:

$$\mu_{t1}(x) = A_{t1}x + B_{t1}, \quad \text{where } A_{t1} = \exp\left(\int_t^1 \alpha(u)\mathrm{d}u\right), \quad B_{t1} = \int_t^1 \exp\left(\int_s^1 \alpha(u)\mathrm{d}u\right)\beta(s)\mathrm{d}s$$

$$\Sigma_{t1} = \left(\int_t^1 \exp\left(2\int_s^1 \alpha(u)\mathrm{d}u\right)\sigma^2(s)\mathrm{d}s\right) I_d$$

By our benchmark design, the exponentiated terminal potential is explicitly parameterized as a Gaussian Mixture Model: $\exp(\psi^*(y)) = \sum_{k=1}^K w_k \mathcal{N}(y;\ \mu_k, \Sigma_k)$. Substituting this and the Gaussian transition kernel R_{t1} into the integral for $h(x,t)$ yields a convolution of two Gaussian densities. Integrating over the terminal state y results in another exact, closed-form Gaussian mixture:

$$h(x,t) = \sum_{k=1}^K w_k \mathcal{N}(A_{t1}x + B_{t1};\ \mu_k, \Sigma_k + \Sigma_{t1}) \tag{4}$$

To obtain the Girsanov drift correction, we compute the spatial gradient of the log-harmonic function, $\nabla_x \log h(x,t) = \frac{\nabla_x h(x,t)}{h(x,t)}$. Applying the chain rule to the mixture in (4), we define the normalized, time-and-state-dependent mixture weights:

$$\hat{w}_k(x,t) = \frac{w_k \mathcal{N}(A_{t1}x + B_{t1};\ \mu_k, \Sigma_k + \Sigma_{t1})}{\sum_{j=1}^K w_j \mathcal{N}(A_{t1}x + B_{t1};\ \mu_j, \Sigma_j + \Sigma_{t1})}$$

This yields the exact, closed-form score correction:

$$\nabla_x \log h(x,t) = -A_{t1} \sum_{k=1}^{K} \hat{w}_k(x,t)(\Sigma_k + \Sigma_{t1})^{-1}(A_{t1}x + B_{t1} - \mu_k)$$

Finally, substituting this score and the linear reference drift back into the general optimal drift equation yields the exact continuous-time optimal vector field for our benchmark:

$$v^*(x,t) = \alpha(t)x + \beta(t) - \sigma^2(t)A_{t1} \sum_{k=1}^{K} \hat{w}_k(x,t)(\Sigma_k + \Sigma_{t1})^{-1}(A_{t1}x + B_{t1} - \mu_k) \qquad (5)$$

3. Evaluation metrics for Schrödinger Bridge solvers

To rigorously assess whether a neural solver has converged to the exact constructed Schrödinger Bridge, we must evaluate the learned model against the ground-truth optimal coupling π^* and the optimal continuous-time drift v^*. Because modern solvers fall into two broad categories—those that learn static couplings or potentials (e.g., via Sinkhorn iterations or parameterized energy-based models) and those that learn the dynamic continuous-time drift (e.g., via forward-backward SDEs)—we divide our evaluation metrics into static and dynamic regimes.

Static Evaluation: Conditional Bures-Wasserstein Metric

For solvers that output a learned conditional plan $\hat{\pi}(\cdot \mid x)$ or directly parameterize the dual potentials to sample terminal states, we measure the divergence between the learned conditional distribution and the ground-truth conditional GMM $\pi^*(\cdot \mid x)$.

Because estimating the exact Kullback-Leibler divergence or continuous Wasserstein distance between arbitrary high-dimensional distributions is computationally intractable, we rely on the conditional Bures-Wasserstein Unexplained Variance Percentage ($c\mathcal{BW}_2^2$-UVP) metric. This metric computes the expected Wasserstein-2 distance between the Gaussian approximations of the conditional plans.

For any initial state $x \sim \mu$, we first compute the exact analytical mean $\mu^*(x)$ and covariance $\Sigma^*(x)$ of our ground-truth conditional GMM $\pi^*(\cdot \mid x)$:

$$\mu^*(x) = \sum_{k=1}^{K} \tilde{w}_k(x) m_k(x)$$

$$\Sigma^*(x) = \sum_{k=1}^{K} \tilde{w}_k(x) \left(S_k + m_k(x) m_k(x)^\top\right) - \mu^*(x)\mu^*(x)^\top$$

Let $\hat{\mu}(x)$ and $\hat{\Sigma}(x)$ denote the empirical mean and covariance estimated from a batch of samples drawn from the learned model $\hat{\pi}(\cdot \mid x)$. The conditional Bures-Wasserstein distance for a given x is:

$$\mathcal{BW}_2^2(\hat{\pi}_x, \pi_x^*) = \|\hat{\mu}(x) - \mu^*(x)\|^2 + \mathrm{Tr}\left(\hat{\Sigma}(x) + \Sigma^*(x) - 2\left(\Sigma^*(x)^{1/2}\hat{\Sigma}(x)\Sigma^*(x)^{1/2}\right)^{1/2}\right)$$

To provide a normalized score across datasets, we integrate this distance over the initial marginal μ and normalize by the variance of the target marginal ν, defining the UVP metric:

$$c\mathcal{BW}_2^2\text{-UVP}(\hat{\pi}, \pi^*) = \frac{100\%}{\frac{1}{2}\mathrm{Tr}(\mathrm{Cov}(\nu))}\int_{\mathcal{X}} \mathcal{BW}_2^2(\hat{\pi}_x, \pi_x^*)\mathrm{d}\mu(x)$$

A score approaching 0% indicates perfect first- and second-moment matching of the optimal conditional plan, while a score of 100% indicates failure equivalent to a naive mapping to the target's global mean.

Dynamic Evaluation: Trajectory-Level Drift Matching

For continuous-time solvers that simulate paths via a learned drift $\hat{v}(x, t)$, evaluating terminal samples is insufficient, as multiple suboptimal drifts can theoretically yield the same terminal marginal. We must evaluate whether the model has learned the true optimal dynamics $v^*(x, t)$ (equation 5) along the entire time horizon $t \in [0, 1]$.

Let T_{v^*} denote the measure of the true Schrödinger Bridge process and $T_{\hat{v}}$ denote the measure of the process governed by the learned drift $\hat{v}(x, t)$. Both processes share the same volatility $\sigma(t)$. By Girsanov's theorem, the Kullback-Leibler divergence between these two path measures is directly proportional to the integrated expected squared difference of their drifts, weighted by the inverse covariance of the noise.

We define two distinct continuous-time metrics: the Forward KL and the Reverse KL.

Forward KL ($\mathcal{L}^2_{fwd}$): Evaluates the learned drift on the ground-truth trajectories. This measures if the neural network has accurately modeled the optimal vector field along the true data manifold:

$$\mathcal{L}^2_{fwd}(t) = \mathbb{E}_{X_t \sim T_{v^*}}\left[\left\|\sigma(t)^{-1}\left(v^*(X_t, t) - \hat{v}(X_t, t)\right)\right\|^2\right]$$
$$\mathrm{KL}(T_{v^*} \mid T_{\hat{v}}) = \frac{1}{2}\int_0^1 \mathcal{L}^2_{fwd}(t)\mathrm{d}t$$

To compute this, we simulate intermediate states X_t using the exact closed-form Gaussian transition kernel of the forward process, and compare the neural output $\hat{v}(X_t, t)$ against the analytical drift $v^*(X_t, t)$ derived in the previous section.

Reverse KL ($\mathcal{L}^2_{rev}$): Evaluates the learned drift on the generated trajectories. This metric exposes compounding errors; if the learned drift pushes the simulated state X_t off the true manifold, the Reverse KL heavily penalizes the model if it cannot recover:

$$\mathcal{L}^2_{rev}(t) = \mathbb{E}_{X_t \sim T_{\hat{v}}} \left[\left\| \sigma(t)^{-1} \left(v^*(X_t, t) - \hat{v}(X_t, t) \right) \right\|^2 \right]$$

$$\mathrm{KL}(T_{\hat{v}} \mid T_{v^*}) = \frac{1}{2} \int_0^1 \mathcal{L}^2_{rev}(t) \mathrm{d}t$$

This is computed by actively simulating trajectories through the SDE solver using the neural drift $\hat{v}$, and querying the analytical ground-truth drift v^* at those generated states.

A solver has successfully converged to the true Schrödinger Bridge if and only if both the Forward KL and Reverse KL approach zero. Discrepancies between the two often indicate out-of-distribution drift failure or compounding simulation errors over time.